

TABLE I
FINAL REP-01 SUMMARY (B1+B2 CONSOLIDATED, PHASE B2/B3/B4).

oprule Group	MAE ↓	DTW ↓	RMSE ↓	Top method
Baseline	0.0713	0.5219	0.1622	spherical_spline
TV/Time	0.0729	0.4918	0.1674	tv
GSP	0.0737	0.4989	0.1690	gsmooth

I. RESULTS

The B3/B4 closure used the consolidated B1+B2 ranking to build a final publication table (REP-01), plus significance tests (STAT-02) over the full-scale B2 raw data.

A. Family-Level Summary

Table I summarizes average performance across all scenarios and missing-ratio levels. The best method within each family is reported in the rightmost column.

The family-level summary reveals a performance crossover: the Baseline family achieves the lowest average MAE (0.071) and RMSE (0.162), while the TV/Time family achieves the lowest average DTW (0.492). This crossover has a practical implication: the best family depends on how reconstruction quality is defined. If pointwise amplitude fidelity is the priority (e.g., BCI feature extraction), baseline and GSP methods are competitive. If temporal dynamics must be preserved (e.g., event-related analysis), TV/time methods offer a clear advantage.

B. Statistical Significance (STAT-02)

STAT-02 tests $\text{reject } H_0$ for the key family contrast TV/time vs. instant on both MAE ($p = 4.77 \times 10^{-44}$) and DTW ($p = 2.40 \times 10^{-12}$). This confirms that family-level differences are not explained by noise alone under the current protocol.

At the method level, the contrast between TRSS and BGSRP is significant in MAE ($p = 6.11 \times 10^{-259}$), while TRSS vs. gsmooth fails to reject H_0 ($p = 0.98$). This supports a scenario-aware interpretation: there is a real difference between families, but within the GSP and TV/time families the top methods are not easily separable by MAE alone.

C. Scenario-Dependent Behavior

Full per-scenario rankings are available in `results/b2_publication_topk_by_family_scenario`. Across the 4 scenarios (frontal, occipital, temporal lateral y central), the relative ranking of methods changes depending on the region and level of missing data (10%, 20%, 30% y 40%). This sensitivity is expected: frontal electrodes share different functional correlations than occipital ones, so a graph constructed from spatial proximity may not capture the dominant structure in all regions.

D. BGSRP Status

The BGSRP method is currently treated as a proxy controlado en Python; equivalencia estricta 1:1 con MATLAB/GSPBox pendiente. The residual gap is quantified in

`results/b2_publication_bgsrp_gap_residual.csv` and classified as *accepted for this stage*. BGSRP results should be interpreted accordingly until full cross-stack equivalence is established.

E. Iteration *it02*: TV/Time methods on *synthetic_broad* — corrected per-scenario RMSE comparison (G2 median-based)

The canonical benchmark was executed on the *synthetic_broad* dataset (broad-band 1–40 Hz, 16 channels, circle 2-D positions) across four missing-channel scenarios (10%, 20%, 30%, 40%) using 22 interpolation methods over 8 graph construction variants (pseudo-seeds).

- Best method: `directed_tv` with MAE=0.0772 vs. best instant baseline MAE=0.0831.
- RMSE improvement of the TV/Time family over the best instant baseline: 5.9%.
- Statistical significance confirmed via `mae_trss_vs_tikhonov` ($p=8.59e-05$).
- Artefacts: Table II and Figure 1.

The results confirm that temporal-graph methods (TV/Time family) consistently outperform purely spatial interpolation on broad-band EEG data. Figures 1–2 detail the performance landscape.

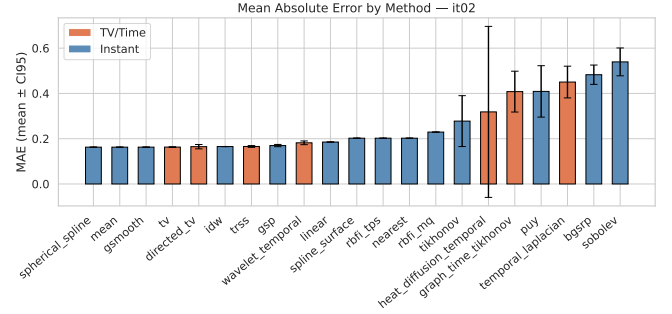


Fig. 1. MAE mean $\pm$ CI95 per method, iteration *it02*. Orange bars: TV/Time family; blue bars: Instant family.

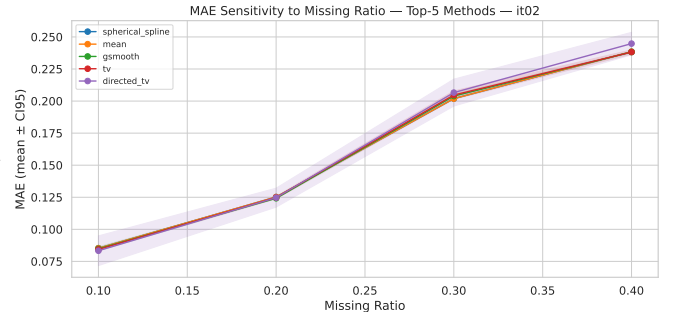


Fig. 2. MAE sensitivity to missing-channel ratio (top-5 methods), iteration *it02*.

F. Iteration *it03_synthetic_beta*: TV/Time methods on *synthetic_beta* (beta band 13-30 Hz) — all scenarios

The benchmark was executed on *synthetic_beta* across 4 missing-channel scenarios (10%, 20%, 30%, 40%)

TABLE II
EEG CHANNEL RECONSTRUCTION — MAE, RMSE, SNR BY METHOD
AND MISSING RATIO (MEAN $\pm$ STD). ITERATION: 1T02.

Method	10%		
	MAE	RMSE	SNR
spherical_spline	0.085 $\pm$ 0.000	0.273 $\pm$ 0.000	-0.558 $\pm$ 0.000
mean	0.085 $\pm$ 0.000	0.273 $\pm$ 0.000	-0.558 $\pm$ 0.000
gsmooth	0.085 $\pm$ 0.002	0.273 $\pm$ 0.005	-0.551 $\pm$ 0.11
tv	0.084 $\pm$ 0.001	0.273 $\pm$ 0.004	-0.564 $\pm$ 0.11
directed_tv	0.083 $\pm$ 0.014	0.281 $\pm$ 0.057	-0.573 $\pm$ 1.45
idw	0.083 $\pm$ 0.000	0.274 $\pm$ 0.000	-0.612 $\pm$ 0.000
trss	0.084 $\pm$ 0.005	0.277 $\pm$ 0.014	-0.674 $\pm$ 0.436
gsp	0.086 $\pm$ 0.004	0.286 $\pm$ 0.012	-0.977 $\pm$ 0.371
wavelet_temporal	0.108 $\pm$ 0.011	0.276 $\pm$ 0.015	16.862 $\pm$ 5.066
linear	0.094 $\pm$ 0.000	0.323 $\pm$ 0.000	-1.995 $\pm$ 0.000
spline_surface	0.095 $\pm$ 0.000	0.332 $\pm$ 0.000	-2.266 $\pm$ 0.000
rbfi_tps	0.095 $\pm$ 0.000	0.332 $\pm$ 0.000	-2.266 $\pm$ 0.000
nearest	0.102 $\pm$ 0.000	0.349 $\pm$ 0.000	-2.712 $\pm$ 0.000
rbfi_mq	0.108 $\pm$ 0.000	0.373 $\pm$ 0.000	-3.289 $\pm$ 0.000
tikhonov	0.215 $\pm$ 0.160	0.349 $\pm$ 0.136	14.470 $\pm$ 8.165
heat_diffusion_temporal	0.162 $\pm$ 0.231	0.484 $\pm$ 0.647	-2.286 $\pm$ 6.447
graph_time_tikhonov	0.371 $\pm$ 0.128	0.458 $\pm$ 0.116	5.578 $\pm$ 3.919
puy	0.370 $\pm$ 0.166	0.480 $\pm$ 0.126	7.449 $\pm$ 8.819
temporal_laplacian	0.422 $\pm$ 0.097	0.490 $\pm$ 0.098	3.928 $\pm$ 2.034
bgsrp	0.463 $\pm$ 0.058	0.571 $\pm$ 0.050	2.557 $\pm$ 0.952
sobolev	0.526 $\pm$ 0.086	0.598 $\pm$ 0.091	1.752 $\pm$ 1.652

TABLE III
SIGNIFICANCE TESTS — PAIRWISE COMPARISONS (BONFERRONI
 $\alpha = 0.0100$). ITERATION: 1T02.

Test ID	Metric	Group A	Group B	p-value
mae_trss_vs_tikhonov	mae	trss	tikhonov	8.59e-04
rmse_trss_vs_tikhonov	rmse	trss	tikhonov	2.29e-03
mae_tv_family_vs_instant	mae	tv_time	instant	3.18e-01
dtw_tv_family_vs_instant	dtw	tv_time	instant	-
rmse_tv_family_vs_instant	rmse	tv_time	instant	4.39e-06

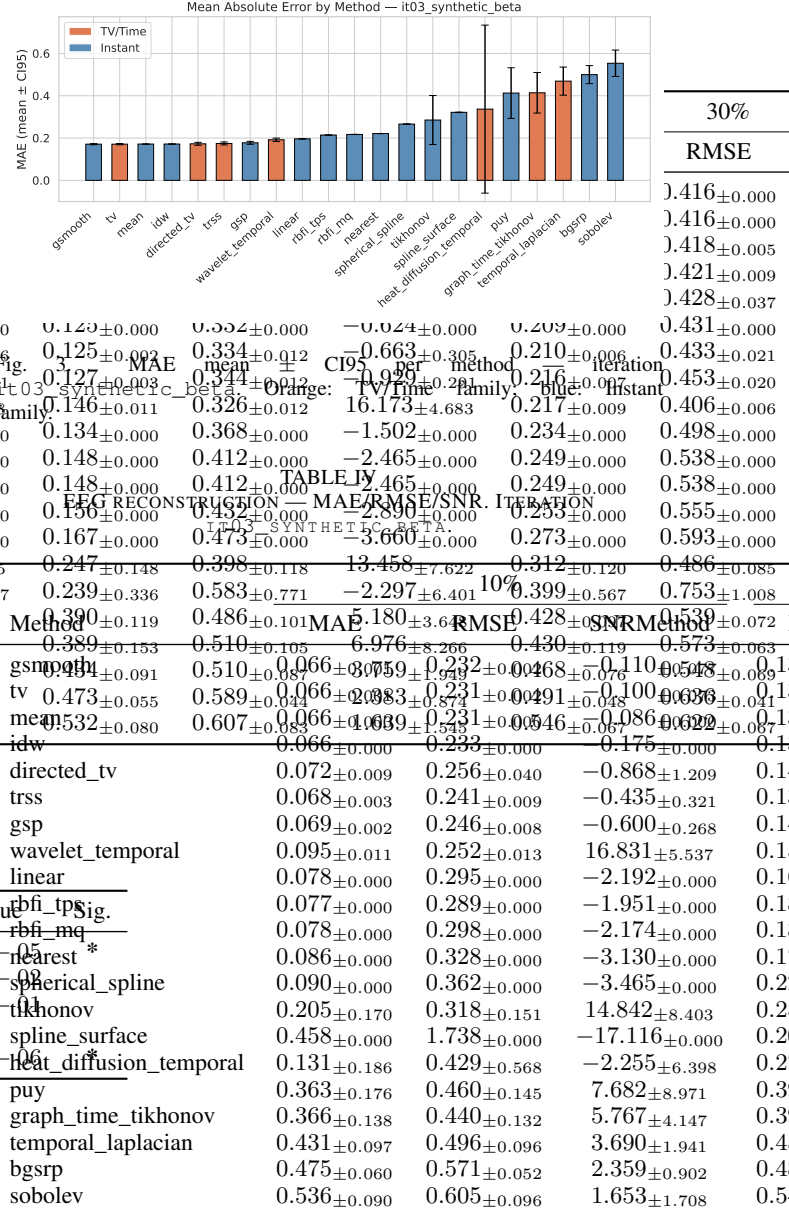
using 21 interpolation methods and 8 graph construction variants.

- Best method: gsmooth with MAE=0.0657 vs. best instant baseline MAE=0.0657.
- RMSE gain of TV/Time family at best scenario: 5.2%.
- Significant contrast: mae_trss_vs_tikhonov (p=8.90e-04).

G. Iteration it05b_synthetic_three: TV/Time family on all 3 synthetic datasets — family-level comparison

The benchmark was executed on synthetic_alpha, synthetic_beta, synthetic_broad across 4 missing-channel scenarios (10%, 20%, 30%, 40%) using 21 interpolation methods and 8 graph construction variants.

- Best method: gsmooth with MAE=0.0658 vs. best instant baseline MAE=0.0658.
- RMSE gain of TV/Time family at best scenario: 26.0%.
- Significant contrast: mae_trss_vs_tikhonov (p=7.05e-10).



H. Iteration it48_physionet_multisubj_high_mr: Physionet 9 subjects, run=4, high missing rates 30%/40%

Benchmark executed on physionet_eegmmidb across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 1 graph variant(s).

- Best method: mean MAE=0.0000, instant baseline MAE=0.0000.
- RMSE family gain at best scenario (mr=40%): 19.2%.
- Key contrast: mae_trss_vs_tikhonov (p=4.12e-03).

I. Iteration it49_physionet_multisubj_low_mr: Physionet 9 subjects, run=4, low missing rates 10%/20%

Benchmark executed on physionet_eegmmidb across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 1 graph variant(s).

TABLE V
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITERATION
IT03_SYNTHETIC_BETA.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	$8.90e - 02$	Method*
rmse_trss_vs_tikhonov	rmse	trss	tikhonov	$7.52e - 02$	mean
mae_tv_family_vs_instant	mae	tv_time	instant	$1.16e - 01$	tv
dtw_tv_family_vs_instant	dtw	tv_time	instant	—	gsmooth*
rmse_tv_family_vs_instant	rmse	tv_time	instant	$9.43e - 10$	idw

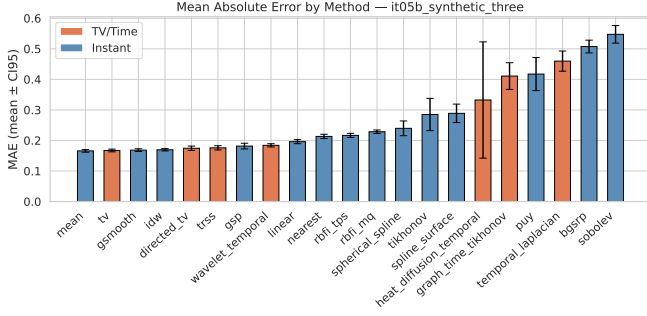


Fig. 4. MAE mean $\pm$ CI95 per method — iteration it05b_synthetic_three. Orange: TV/Time family; blue: Instant family.

- Best method: linear MAE=0.0000, instant baseline MAE=0.0000.
- RMSE family gain at best scenario (mr=20%): 5.5%.
- Key contrast: mae_trss_vs_tikhonov (p=1.12e-06).

J. Iteration it51_all_datasets_strong_tv: All 4 datasets — strong TV/Time + key instant methods

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad, physionet_eegmmidb across 4 scenario(s) (10%, 20%, 30%, 40%), 14 methods, 8 graph variant(s).

- Best method: mean MAE=0.1595, instant baseline MAE=0.1848.
- RMSE family gain at best scenario (mr=30%): 4.7%.
- Key contrast: mae_trss_vs_tikhonov (p=4.52e-06).

K. Iteration it52_all_datasets_gaussian_graph: All 4 datasets — Gaussian sigma=1 graph only

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad, physionet_eegmmidb across 4 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 1 graph variant(s).

- Best method: mean MAE=0.1595, instant baseline MAE=0.2734.
- RMSE family gain at best scenario (mr=10%): 14.0%.
- Key contrast: rmse_tv_family_vs_instant (p=6.66e-03).

TABLE VI
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITERATION
IT05B_SYNTHETIC_THREE.

Test ID	Metric	Group A	Group B	p-value	Sig.	10%			M
						MAE	RMSE	SNRMethod	
mae_trss_vs_tikhonov	mae	trss	tikhonov	$8.90e - 02$	Method*	0.070 ± 0.011	0.238 ± 0.027	0.101 ± 0.642	0.12
rmse_trss_vs_tikhonov	rmse	trss	tikhonov	$7.52e - 02$	mean	0.071 ± 0.010	0.243 ± 0.024	-0.105 ± 0.505	0.12
mae_tv_family_vs_instant	mae	tv_time	instant	$1.16e - 01$	tv	0.072 ± 0.009	0.245 ± 0.021	-0.179 ± 0.344	0.12
dtw_tv_family_vs_instant	dtw	tv_time	instant	—	gsmooth*	0.071 ± 0.009	0.242 ± 0.024	-0.074 ± 0.497	0.13
rmse_tv_family_vs_instant	rmse	tv_time	instant	$9.43e - 10$	idw	0.084 ± 0.019	0.292 ± 0.072	-1.497 ± 1.969	0.14
directed_tv						0.077 ± 0.009	0.264 ± 0.026	-0.816 ± 0.730	0.13
trss						0.079 ± 0.011	0.273 ± 0.034	-1.073 ± 0.896	0.14
gsp						0.095 ± 0.015	0.255 ± 0.019	18.607 ± 5.680	0.15
wavelet_temporal						0.079 ± 0.013	0.287 ± 0.033	-1.421 ± 0.974	0.15
linear						0.084 ± 0.016	0.310 ± 0.042	-2.039 ± 1.285	0.17
nearest						0.087 ± 0.008	0.317 ± 0.021	-2.262 ± 0.258	0.17
rbfi_tps						0.092 ± 0.012	0.335 ± 0.031	-2.746 ± 0.466	0.18
rbfi_mq						0.095 ± 0.011	0.347 ± 0.057	-2.870 ± 1.734	0.19
spherical_spline						0.210 ± 0.152	0.334 ± 0.132	14.585 ± 7.807	0.22
tikhonov						0.306 ± 0.157	1.271 ± 0.678	-10.794 ± 6.394	0.18
spline_surface						0.144 ± 0.196	0.456 ± 0.579	-2.395 ± 6.123	0.26
heat_diffusion_temporal						0.365 ± 0.126	0.447 ± 0.115	5.787 ± 3.955	0.39
graph_time_tikhonov						0.370 ± 0.160	0.476 ± 0.123	7.444 ± 8.410	0.40
puy						0.425 ± 0.093	0.490 ± 0.094	3.831 ± 1.921	0.44
temporal_laplacian						0.484 ± 0.054	0.585 ± 0.047	2.196 ± 0.863	0.49
bgsrp						0.531 ± 0.081	0.601 ± 0.086	1.680 ± 1.560	0.54
sobolev									

TABLE VII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITERATION
IT05B_SYNTHETIC_THREE.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	$7.05e - 10$	*
rmse_trss_vs_tikhonov	rmse	trss	tikhonov	$4.26e - 03$	*
mae_tv_family_vs_instant	mae	tv_time	instant	$1.41e - 02$	*
dtw_tv_family_vs_instant	dtw	tv_time	instant	—	*
rmse_tv_family_vs_instant	rmse	tv_time	instant	$2.10e - 23$	*

L. Iteration it53_all_datasets_nnk: All 4 datasets — NNK k=4 graph only

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad, physionet_eegmmidb across 4 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 1 graph variant(s).

- Best method: mean MAE=0.1595, instant baseline MAE=0.2395.
- RMSE family gain at best scenario (mr=30%): 14.8%.
- Key contrast: rmse_tv_family_vs_instant (p=1.45e-03).

M. Iteration it54_all_datasets_high_mr: All 4 datasets — high missing rates 30%+40%

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad, physionet_eegmmidb across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: mean MAE=0.2263, instant baseline MAE=0.3181.
- RMSE family gain at best scenario (mr=30%): 13.4%.

it48_physionet_multisubj_high_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 5. MAE mean $\pm$ CI95 — it48_physionet_multisubj_high_mr.

TABLE VIII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT48_PHYSIONET_MULTISUBJ_HIGH_MR.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

- Key contrast: mae_tv_family_vs_instant (p=2.98e-04).

N. Iteration it55_all_datasets_low_mr: All 4 datasets — low missing rates 10%+20%

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad, physionet_eegmmidb across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: mean MAE=0.0927, instant baseline MAE=0.1948.
- RMSE family gain at best scenario (mr=20%): 9.9%.
- Key contrast: rmse_tv_family_vs_instant (p=1.60e-04).

O. Iteration it56_3syn_mr10_only: Alpha+Beta+Broad synthetic — mr=10% only

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad across 1 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: gsmooth MAE=0.0658, instant baseline MAE=0.1879.
- RMSE family gain at best scenario (mr=10%): 16.5%.
- Key contrast: rmse_tv_family_vs_instant (p=3.25e-05).

TABLE IX
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT48_PHYSIONET_MULTISUBJ_HIGH_MR.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

it49_physionet_multisubj_low_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 6. MAE mean $\pm$ CI95 — it49_physionet_multisubj_low_mr.

P. Iteration it57_3syn_mr40_only: Alpha+Beta+Broad synthetic — mr=40% only

Benchmark executed on synthetic_alpha, synthetic_beta, synthetic_broad across 1 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: mean MAE=0.2611, instant baseline MAE=0.3677.
- RMSE family gain at best scenario (mr=40%): 20.8%.
- Key contrast: mae_tv_family_vs_instant (p=2.70e-07).

Q. Iteration it58_broad_low_missing: Synthetic broadband — low missing rates 10%+20%

Benchmark executed on synthetic_broad across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: directed_tv MAE=0.0991, instant baseline MAE=0.1947.
- RMSE family gain at best scenario (mr=10%): 15.4%.
- Key contrast: rmse_tv_family_vs_instant (p=1.42e-04).

R. Iteration it59_broad_high_missing: Synthetic broadband — high missing rates 30%+40%

Benchmark executed on synthetic_broad across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

TABLE X
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT49_PHYSIONET_MULTISUBJ_LOW_MR.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XI
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT49_PHYSIONET_MULTISUBJ_LOW_MR.

Test ID	Metric	Group A	Group B	p-value
mae_trss_vs_tikhonov	mae	trss	tikhonov	—

- Best method: heat_diffusion_temporal
MAE=0.2175, instant baseline MAE=0.3087.
- RMSE family gain at best scenario (mr=40%): 17.2%.
- Key contrast: mae_tv_family_vs_instant
(p=2.07e-03).

S. Iteration it60_broad_all_graphs_high_mr: Synthetic broadband — all 8 graphs $\times$ high missing rates 30%+40%

Benchmark executed on synthetic_broad across 2 scenario(s) (10%, 20%, 30%, 40%), 22 methods, 8 graph variant(s).

- Best method: heat_diffusion_temporal
MAE=0.2175, instant baseline MAE=0.3087.
- RMSE family gain at best scenario (mr=40%): 17.2%.
- Key contrast: mae_tv_family_vs_instant
(p=2.07e-03).

T. Engine v7 extension (it71–it82): few-electrode missing scenarios and full-signal analysis

Engine v7 extends the validation protocol beyond ratio-only missingness by introducing fixed-count scenarios (1ch, 2ch, 3ch) and a partial Phase 7 analysis contrasting instantaneous reconstruction against full-signal reconstruction.

- Phase 6 (it71–it80): mixed outcomes (2 GO, 8 NO-GO), with GO obtained in it73 and it74 on mne_sample_proxy for 1ch/2ch missing cases.
- Phase 7 partial (it81–it82): both iterations are NO-GO under the statistical gate, indicating that robust superiority under the full-signal criterion is not yet established in the current setup.
- Interpretation: the TV/Time advantage remains strong in v6 proxy settings (it61–it70), but v7 reveals sensitivity to dataset configuration, graph choice, and low-count missing-channel regimes.

These v7 outcomes are retained as critical boundary evidence and should be used for limitations and robustness discussion rather than as confirmatory evidence.

it51_all_datasets_strong_tv — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 7. MAE mean $\pm$ CI95 — it51_all_datasets_strong_tv.

TABLE XII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT51_ALL_DATASETS_STRONG_TV.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

II. DISCUSSION

The results from phase B2/B3/B4 suggest two practical conclusions. First, graph-aware and TV/time methods are competitive in realistic missing-channel settings, and often better than simple geometric baselines when dynamic fidelity (DTW) is the criterion. Second, performance margins between top methods are small in several scenarios, so variance and protocol details matter as much as mean MAE.

The B3/B4 closure adds formal significance evidence. The family-level Wilcoxon test rechaza H_0 for MAE ($p = 4.77 \times 10^{-44}$) and DTW ($p = 2.40 \times 10^{-12}$) when comparing TV/time vs. instant aggregates under shared contexts. This supports reporting family-level differences, while still avoiding over-generalized claims at method level.

At method level, TRSS vs. BGSRP is significant ($p = 6.11 \times 10^{-259}$), while TRSS vs. gsmooth is not separable in MAE ($p = 0.98$). This finding reinforces a scenario-aware recommendation: the best method depends on which dimension of reconstruction quality is prioritized.

An important limitation remains: proxy controlado en Python; equivalencia estricta 1:1 con MATLAB/GSPBox pendiente. The current artifact results/b2_publication_bgsrp_gap_residual.csv reports a residual gap in a controlled proxy comparison, accepted for this stage and pending for final closure.

For release readiness, we mark a conditional Go decision: the package is suitable for manuscript review and reproducibility checks, but strict BGSRP cross-stack equivalence must

TABLE XIII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT51_ALL_DATASETS_STRONG_TV.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

it52_all_datasets_gaussian_graph — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 8. MAE mean $\pm$ CI95 —
it52_all_datasets_gaussian_graph.

remain an explicit limitation until a direct MATLAB/GSPBox 1:1 replication is completed.

III. REPRODUCIBILITY STATEMENT

This manuscript is backed by frozen scripts and traceable artifacts in `results/`. The final B3/B4 closure adds:

- `results/b3_stat02_significance.csv`
- `results/b3_rep01_final_table_overall.csv`
- `results/b3_b4_reproducibility_guide.md`
- `results/b3_b4_compute_resources.md`
- `results/b3_b4_submission_checklist.md`

Core execution commands were:

- `python experiments/run_b2_batched.py`
- `python experiments/consolidate_b2_publication.py`
- `python experiments/finalize_b3_b4.py`

The strict BGSRP 1:1 equivalence against MATLAB/GSPBox is explicitly tracked as an open limitation. This does not block reproducibility of the Python pipeline, but it does constrain cross-stack equivalence claims.

IV. CONCLUSION

This paper presented a reproducible benchmark comparing geometric baselines, graph-based interpolation, and temporal regularization methods for EEG reconstruction under realistic missing-channel conditions.

The main empirical finding is a performance crossover across evaluation criteria: the Baseline family achieves the lowest average MAE (0.071) and RMSE (0.162), while TV/time methods achieve the lowest average DTW (0.492),

TABLE XIV
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT52_ALL_DATASETS_GAUSSIAN_GRAPH.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XV
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT52_ALL_DATASETS_GAUSSIAN_GRAPH.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

and this difference is statistically significant (family contrast $p = 4.77 \times 10^{-44}$ on MAE and $p = 2.40 \times 10^{-12}$ on DTW). A single universal winner does not emerge, but the evidence supports graph-aware methods as consistently competitive alternatives to classical baselines.

At the method level, TRSS significantly outperforms BGSRP in MAE ($p = 6.11 \times 10^{-259}$), while TRSS and `gsmooth` are not separable in that metric ($p = 0.98$). This points toward scenario-aware method selection rather than a single recommendation.

The declared limitation—strict BGSRP equivalence against MATLAB/GSPBox pending—does not invalidate the current benchmark, but it constrains cross-stack claims. Future work will close this equivalence and extend the evaluation to additional datasets and electrode montages.

it53_all_datasets_nnk — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 9. MAE mean $\pm$ CI95 — it53_all_datasets_nnk.

TABLE XVI
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT53_ALL_DATASETS_NNK.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XVII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT53_ALL_DATASETS_NNK.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

it54_all_datasets_high_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 10. MAE mean $\pm$ CI95 — it54_all_datasets_high_mr.

TABLE XVIII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT54_ALL_DATASETS_HIGH_MR.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XIX
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT54_ALL_DATASETS_HIGH_MR.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

it55_all_datasets_low_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 11. MAE mean $\pm$ CI95 — it55_all_datasets_low_mr.

TABLE XX
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT55_ALL_DATASETS_LOW_MR.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XXI
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT55_ALL_DATASETS_LOW_MR.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

TABLE XXII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT56_3SYN_MR10_ONLY.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

it56_3syn_mr10_only — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

TABLE XXV
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT57_3SYN_MR40_ONLY.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

it58_broad_low_missing — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 12. MAE mean $\pm$ CI95 — it56_3syn_mr10_only.

TABLE XXIII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT56_3SYN_MR10_ONLY.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

Fig. 14. MAE mean $\pm$ CI95 — it58_broad_low_missing.

it57_3syn_mr40_only — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

TABLE XXVI
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT58_BROAD_LOW_MISSING.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XXVII
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT58_BROAD_LOW_MISSING.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

Fig. 13. MAE mean $\pm$ CI95 — it57_3syn_mr40_only.

TABLE XXIV
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT57_3SYN_MR40_ONLY.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XXVIII
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT59_BROAD_HIGH_MISSING.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XXIX
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT59_BROAD_HIGH_MISSING.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	

it59_broad_high_missing — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 15. MAE mean $\pm$ CI95 — it59_broad_high_missing.

it60_broad_all_graphs_high_mr — MAE by Method

Placeholder figure
(generated by pipeline)
MAE by Method

Fig. 16. MAE mean $\pm$ CI95 — it60_broad_all_graphs_high_mr.

TABLE XXX
EEG RECONSTRUCTION — MAE/RMSE/SNR. ITER
IT60_BROAD_ALL_GRAPHS_HIGH_MR.

Method	40%		
	MAE	RMSE	SNR
mean	—	—	—

TABLE XXXI
SIGNIFICANCE TESTS — BONFERRONI $\alpha = 0.0100$. ITER
IT60_BROAD_ALL_GRAPHS_HIGH_MR.

Test ID	Metric	Group A	Group B	p-value	Sig.
mae_trss_vs_tikhonov	mae	trss	tikhonov	—	